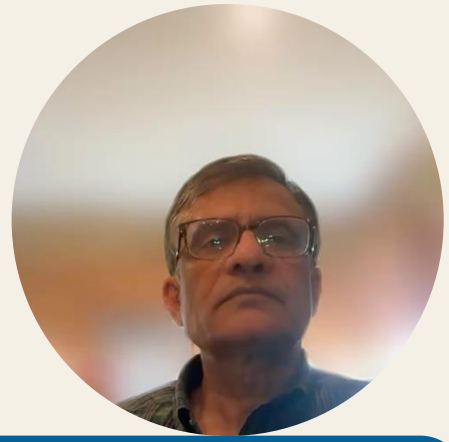
Embedded System HW Design
Week3
Selecting Microcontrollers

- Selecting Microcontrollers
- Compare the different type of processor architectures
- ARM Cortex-M architecture



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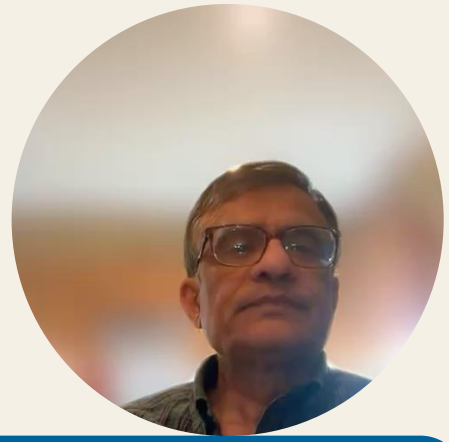
Embedded System HW Design
Week3
Why MCU Selection Matters

- MCU choice impacts **cost, power, performance, schedule**

- Wrong choice → redesign risk, integration delays

- MCU drives: peripherals, memory, toolchain, firmware structure



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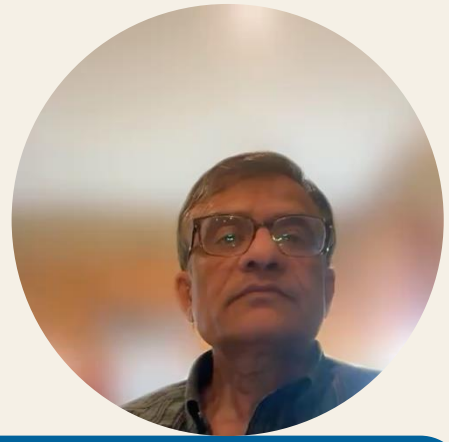
Embedded System HW Design
Week3
MCU vs MPU vs SoC (High-Level)

- **MCU:** integrated CPU + Flash + RAM + peripherals

- **MPU:** higher performance, often runs Linux, external memory

- **SoC:** integrated complex system, multimedia/AI, high integration



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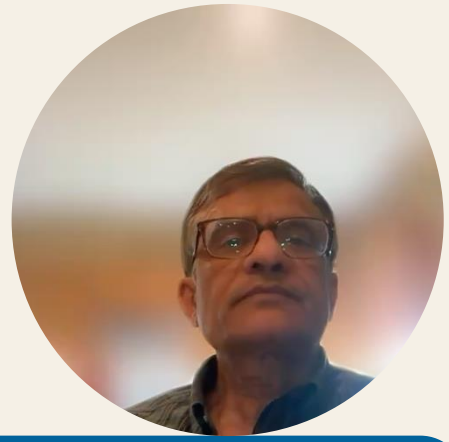
Embedded System HW Design
Week3
Start With Requirements

- Functional requirements (what it must do)
- Non-functional (power, cost, size, boot time, reliability)
- Real-world constraints: availability, lifecycle, vendor support



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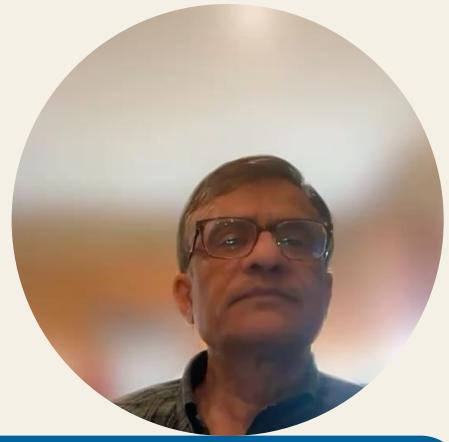
Embedded System HW Design
Week3
MCU Selection Checklist

- The “top 10” specs engineers check:
- Core type + frequency
- Flash / RAM
- Peripherals (I^2C/SPI/UART/USB/CAN)
- ADC/DAC + timers/PWM
- GPIO count + pin mux limits
- Power modes & current draw
- Voltage range
- Package & temperature grade
- Debug support (SWD/JTAG)
- Toolchain + ecosystem



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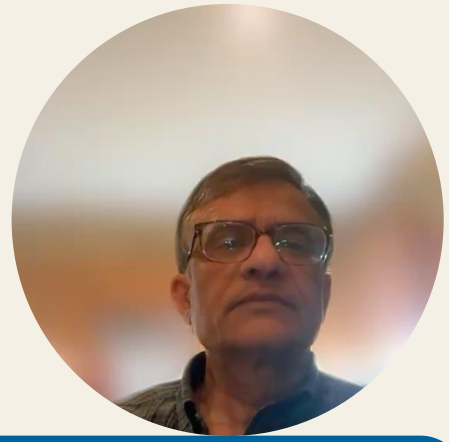
Embedded System HW Design
Week3
CPU Core Choice (Cortex-M Overview)

- M0/M0+ (ultra-low power / simple)
- M3/M4 (general + DSP)
- M7 (high performance)
- M33 (security + low power)
- FPU vs non-FPU



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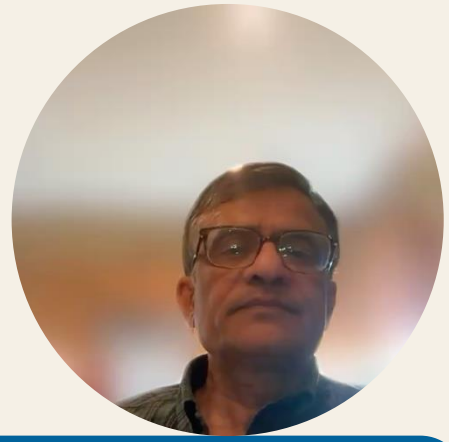
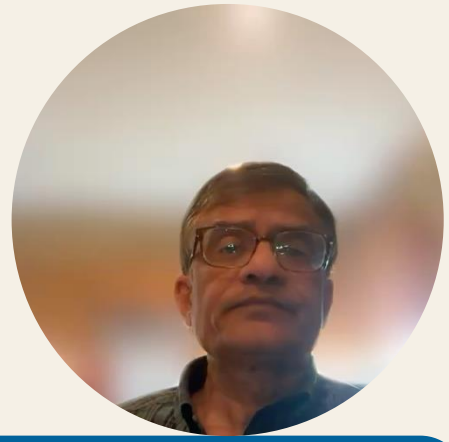
Embedded System HW Design
Week3
Memory Sizing Rules of Thumb

- Flash sizing (code + libraries + updates)
- RAM sizing (buffers, stacks, RTOS tasks)
- Headroom planning (20–40% margin)
- External Flash / SD when needed



Embedded System HW Design
Week3
Peripheral Fit: What Matters Most

- Examples:
- Sensors → I^2C / SPI
- Motor control → PWM timers + ADC + interrupts
- Logging → QSPI / SDIO
- Connectivity → UART, USB, SPI + modules
- Real-time → timer resolution & interrupt latency



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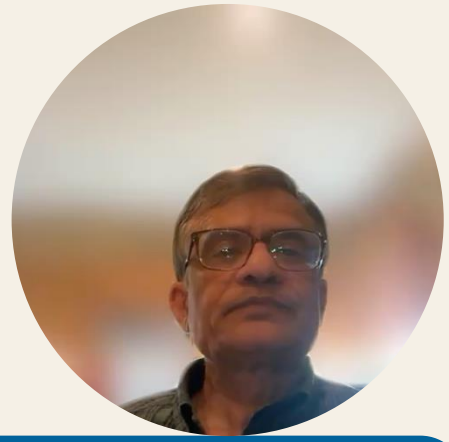
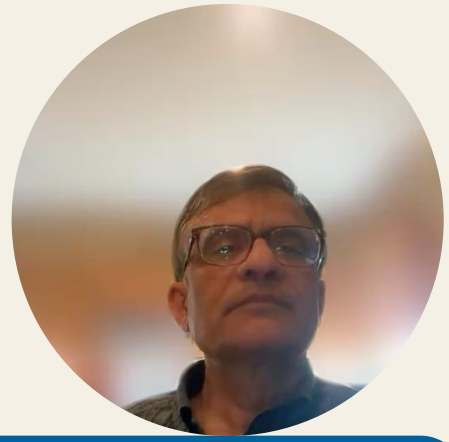
Embedded System HW Design
Week3
Pins & Packaging Pitfalls

- Pin mux conflicts (you can’t map everything everywhere)
- GPIO count vs real usable pins
- Package choices: QFN vs LQFP vs BGA
- PCB complexity tradeoff



Embedded System HW Design
Week3
Power & Battery Life

- Active current vs sleep current
- Clock speed vs power
- Peripheral power dominates sometimes
- Low-power modes strategy



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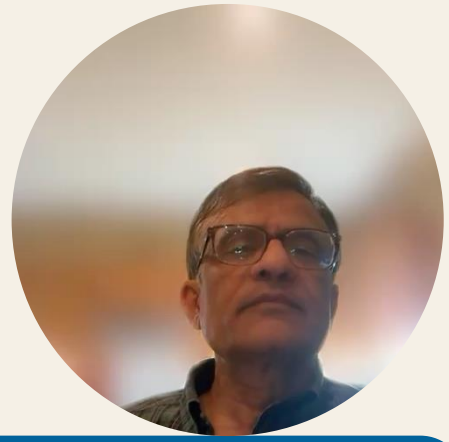
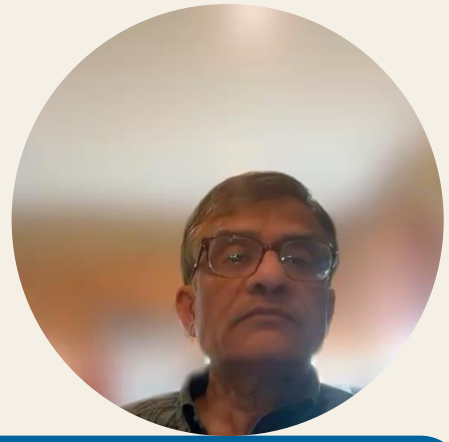
Embedded System HW Design
Week3
Cost + Supply Chain Reality

- Unit cost vs total system cost
- Lead times, multi-source options
- Lifecycle status (NRND / EOL risk)
- “Cheaper MCU” can increase BOM cost elsewhere



Embedded System HW Design
Week3
Development Ecosystem

- IDE/toolchain quality
- HAL vs registers vs libraries
- Middleware availability (USB, BLE stacks)
- Community + examples



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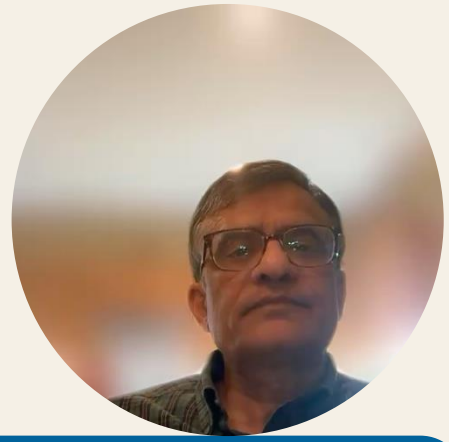
Embedded System HW Design
Week3
COTS Module + MCU Pairing

- When MCU choice depends on module interface
- Example: Wi-Fi module uses UART/SPI
- Memory/performance considerations for protocol stack



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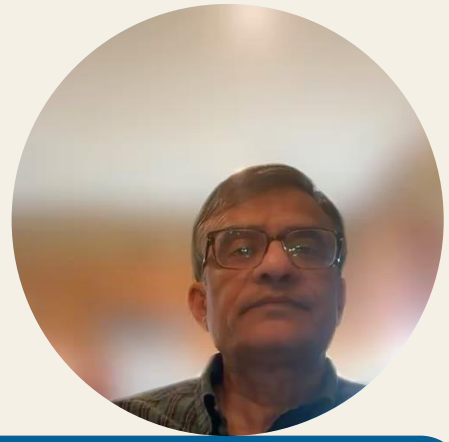
Embedded System HW Design
Week3
Example Selection: STM32L475 (Discovery Board)

- Use your board as a real case study:
- Cortex-M4 + FPU
- Low-power design focus
- Rich sensor set & connectivity
- Why it’s a good teaching platform



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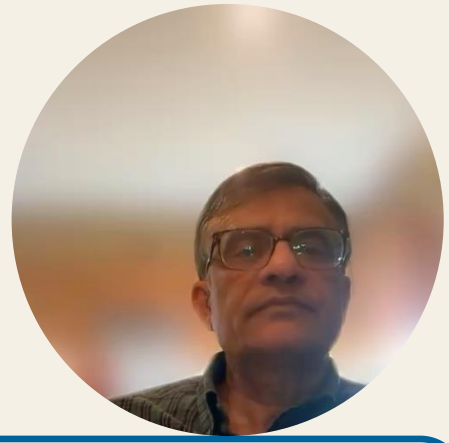
Embedded System HW Design
Week3
Decision Matrix Slide (Scoring Table)

Candidate MCU	Cost	Power	Performance	Memory	Peripherals	Tooling	Risk
MCU Option A							
MCU Option B							
MCU Option C							
MCU Option D (optional)							



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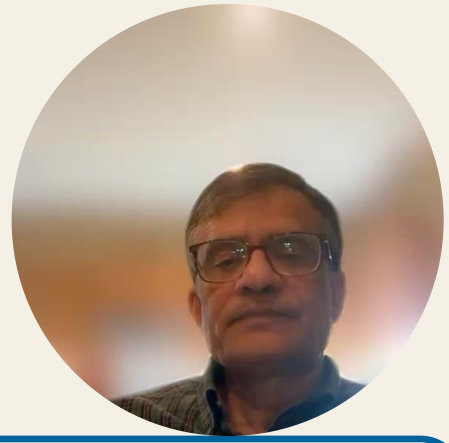
Embedded System HW Design
Week3
Common Mistakes

- Choosing MCU too early
- Underestimating RAM
- Ignoring power modes
- Pin mux conflicts
- Assuming libraries are “free”



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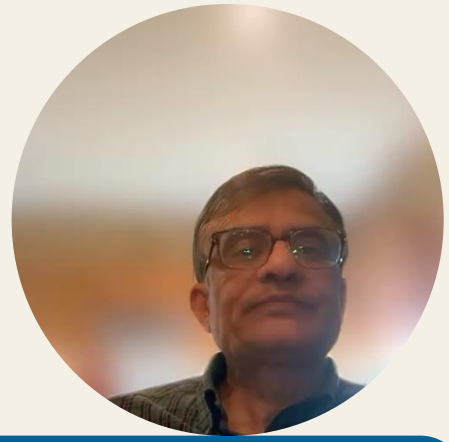
Embedded System HW Design
Week3
Summary & Key Takeaway

- “The best MCU isn’t the most powerful one.

It’s the one that meets requirements with margin, lowest risk, and best fit.”



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Embedded System HW Design
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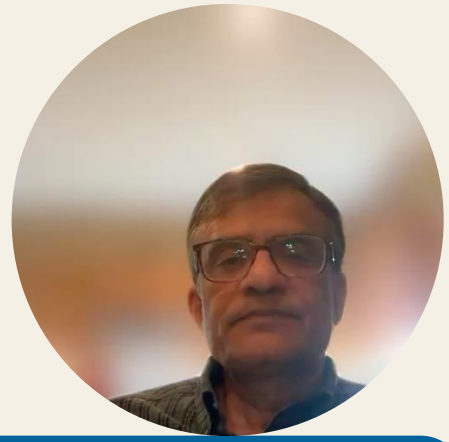

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